DATA STRUCTURE

LAB PROGRAMS

1. a) Write a C program to perform Matrix Multiplication using 2D.

```
#include <stdio.h>
int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r1, c1, r2, c2, i, j, k;
    printf("Enter rows and columns of first matrix: ");
    scanf("%d %d", &r1, &c1);
    printf("Enter rows and columns of second matrix: ");
    scanf("%d %d", &r2, &c2);
    if (c1 != r2)
    {
        printf("Matrix multiplication is not possible.\n");
        return 0;
    }
    printf("Enter elements of first matrix:\n");
    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c1; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
```

```
printf("Enter elements of second matrix:\n");
for(i = 0; i < r2; i++)
{
   for(j = 0; j < c2; j++)
   {
      scanf("%d", &b[i][j]);
   }
}
for(i = 0; i < r1; i++)
{
   for(j = 0; j < c2; j++)
   {
      c[i][j] = 0;
      for(k = 0; k < c1; k++)
      {
         c[i][j] += a[i][k] * b[k][j];
      }
   }
}
printf("Resultant Matrix:\n");
for(i = 0; i < r1; i++)
{
   for(j = 0; j < c2; j++)
   {
      printf("%d\t", c[i][j]);
   }
   printf("\n");
}
```

```
    return 0;
}
```

```
Output
Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2
Enter elements of first matrix:
1 2
3 4
Enter elements of second matrix:
5 6
7 8
Resultant Matrix:
19  22
43  50


=== Code Execution Successful ===
```

b) Write a C program to perform Matrix Multiplication using 3D.

```
#include <stdio.h>
#define DEPTH 2
#define ROWS 3
#define COLS 3
void multiply3DMatrices(int A[DEPTH][ROWS][COLS],
                        int B[DEPTH][ROWS][COLS],
                        int C[DEPTH][ROWS][COLS])
{
    for(int d = 0; d < DEPTH; d++)
    {
```

```
        for(int i = 0; i < ROWS; i++)
        {
            for(int j = 0; j < COLS; j++)
            {
                C[d][i][j] = 0;
                for(int k = 0; k < COLS; k++)
                {
                    C[d][i][j] += A[d][i][k] * B[d][k][j];
                }
            }
        }
    }
}
int main()
{
    int A[DEPTH][ROWS][COLS] =
    {
        {
            {1,2,3},
            {4,5,6},
            {7,8,9}
        },
        {
            {9,8,7},
            {6,5,4},
            {3,2,1}
        }
    };
    int B[DEPTH][ROWS][COLS] =
```

```
    {
        {
            {9,8,7},
            {6,5,4},
            {3,2,1}
        },
        {
            {1,2,3},
            {4,5,6},
            {7,8,9}
        }
    };
    int C[DEPTH][ROWS][COLS] = {0};
    multiply3DMatrices(A, B, C);
    printf("Result:\n");
    for(int d = 0; d < DEPTH; d++)
    {
        printf("\nLayer %d:\n", d + 1);
        for(int i = 0; i < ROWS; i++)
        {
            for(int j = 0; j < COLS; j++)
            {
                printf("%d\t", C[d][i][j]);
            }
            printf("\n");
        }
    }
    return 0;
}
```

Output

```
Result:

Layer 1:
30  24  18
84  69  54
138 114 90

Layer 2:
90  114 138
54  69  84
18  24  30


=== Code Execution Successful ===
```